
AWS KMS — Key Features

1. Centralized Key Management

AWS KMS lets you **create, manage, and control cryptographic keys** in one central place.

 Use it across services like:

- S3
- EBS
- RDS
- Lambda
- CloudTrail, and more!

2. Envelope Encryption

KMS uses **envelope encryption** to efficiently encrypt data:

- A **data key** encrypts the actual data.
- A **CMK** (Customer Master Key) encrypts the data key.

 This provides speed, security, and scalability.

3. Customer Managed Keys (CMKs)

You can create and manage your own encryption keys:

- Define key policies

- Enable/disable keys
- Control access with IAM

👑 You have **full ownership and control**.

4. AWS Managed Keys

AWS automatically manages encryption keys for you:

- One key per service per account
- Simplified encryption setup

🔧 Example: `aws/s3` for S3 server-side encryption (SSE-KMS)

5. Audit and Logging with AWS CloudTrail

KMS integrates with **AWS CloudTrail** to log:

- Who used the key
- When it was used
- Which service used it

📜 Great for **compliance**, **security auditing**, and **forensics**.

6. Automatic Key Rotation

For Customer Managed Keys (CMKs), KMS can automatically:

- Rotate keys **once every 365 days** 
- Maintain old versions for decryption

🧠 Ensures long-term security with minimal effort.

🔑 7. Granular Access Control

Define access with:

- **Key policies**
- **IAM policies**
- **Grants** (temporary permissions)

🎯 You control **who can use, manage, or rotate** each key.

🧩 8. Integrated with AWS Services

KMS is natively integrated with 70+ AWS services, like:

- 📦 S3 (bucket/object encryption)
- 📊 RDS (database encryption)
- 💾 EBS (volume encryption)
- 🛠️ Lambda (env vars encryption)
- 🔍 CloudTrail (log encryption)

🔗 Encryption becomes **plug-and-play**.

🌐 9. Global and Multi-Region Support

- Create **multi-Region keys** 🌐
- Easily replicate CMKs across regions

- Simplifies disaster recovery and latency-sensitive apps
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10. Key Aliases

You can create **aliases** for easier reference.

 Example:

alias/my-app-key

 Use it instead of a long Key ID.

11. FIPS 140-2 Compliance

KMS supports **FIPS 140-2** validated cryptographic modules — suitable for highly regulated industries.

 Great for:

- Healthcare
 - Financial services
 - Government use cases
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12. Custom Key Store (CKS)

For highly secure environments:

- Store your CMKs in **dedicated HSMs** using AWS CloudHSM
- Full control over key storage and lifecycle

 For advanced compliance and isolation requirements.

13. Usage Tracking and Metrics

Monitor KMS usage with **CloudWatch metrics**, including:

- Key usage count
- Errors
- Throttles

 Helps identify unusual activity or misconfigurations.

14. Key Deletion and Recovery

- Schedule key deletion (7–30 days waiting period)
- Gives you time to **cancel deletion** if needed

 Once deleted, **data encrypted with the key becomes unrecoverable!**

Summary Table

Feature 	Description 
 Key Management	Create & manage keys easily
 Rotation	Automatic yearly rotation
 AWS Integration	Plug-and-play with AWS services
 Auditing	Track all key usage via CloudTrail
 Access Control	IAM + Key policies + Grants
 Compliance	FIPS, HIPAA, PCI-DSS support
 Multi-Region	Resilient architecture
 Custom HSM Store	High-security use cases



CloudWatch Metrics Monitor key activity
